

Technical report: Exact Maximum Flexi-Clique Computation under Non-Hereditary Feasibility and Connectivity Constraints

A Hardness proof of maximum Flexi-clique problem

A.1 NP-hardness

LEMMA 1. *The Flexi-clique decision problem is NP-hard.*

PROOF. We prove NP-hardness by reduction from the well-known k -clique problem, which is NP-complete in its decision form. Given an instance (G', k) of the clique decision problem, we construct an equivalent instance (G, τ, k) of the Flexi-clique decision problem as follows. We set $G = G'$, meaning that the node and edge sets remain unchanged. We then choose $\tau \in (0, 1)$ sufficiently close to 1 so that for any subset of size x , the degree threshold satisfies $\lfloor x^\tau \rfloor = x - 1$. Intuitively, this setting makes the Flexi-clique condition equivalent to the clique constraint. Formally, it suffices to select τ satisfying

$$\frac{\ln(|V| - 1)}{\ln(|V|)} < \tau < 1,$$

since for any $x < x'$, the ratio $\frac{\ln(x-1)}{\ln(x)} < \frac{\ln(x'-1)}{\ln(x')}$. The parameter k remains the same. If G' contains a clique H of size at least k , then H also forms a valid Flexi-clique in G , because each node has degree $|H| - 1$ within H . Conversely, if G contains a Flexi-clique H of size at least k , the degree constraint forces H to be a clique, implying that G' contains a clique of size at least k . Since the transformation runs in polynomial time, the Flexi-clique decision problem is NP-hard. $\square$

LEMMA 2. *The Maximum Flexi-clique problem is NP-hard.*

PROOF. We prove NP-hardness by reducing from the Flexi-clique decision problem (shown to be NP-hard in Lemma 1). Suppose we have a polynomial-time algorithm $\mathcal{A}$ that finds the maximum Flexi-clique in any instance (G, τ) . Given an instance (G, τ, k) of the k -Flexi-clique problem, we:

- (1) Use $\mathcal{A}$ to compute the largest Flexi-clique $H_{\max}$ and its size $|H_{\max}|$.
- (2) If $|H_{\max}| \geq k$, return yes; otherwise, return no.

Since it solves the NP-hard k -Flexi-clique problem in polynomial time, the Maximum Flexi-clique problem must also be NP-hard. $\square$

A.2 Constant factor inapproximability

THEOREM 1. *The Maximum Flexi-clique problem is constant-factor inapproximable unless $P = NP$.*

PROOF. We prove the claim via a gap-preserving reduction from the gap version of the CLIQUE problem. Specifically, given a graph $G = (V, E)$, an integer K , and a constant $c > 1$, the task is to distinguish between the following two cases, under the promise that one of them holds:

- **YES:** G contains a clique of size at least K ;
- **NO:** every clique in G has size at most K/c .

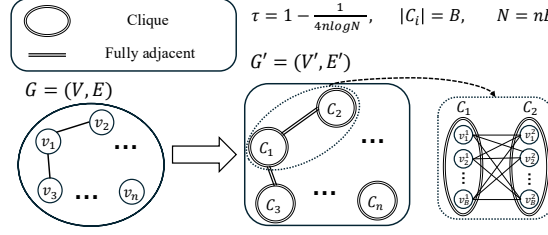


Fig. 1. Reduction process

It is NP-hard to distinguish between the above two cases [1]. Given a graph $G = (V, E)$ with parameters K and $c > 1$, under the promise that G contains a clique of size at least K or that every clique in G has size at most K/c , we construct a graph $G' = (V', E')$ that preserves the promised gap while amplifying differences in clique size. Specifically, for each node $v \in V$, we create a clique C_v consisting of exactly B nodes, where B is a sufficiently large polynomial in n . The node set of G' is defined as $V' = \bigcup_{v \in V} C_v$, and we denote $|V'|$ by N . For any two distinct nodes $u, v \in V$, the cliques C_u and C_v are made fully adjacent if $(u, v) \in E$, and completely non-adjacent otherwise. Next, we set the parameter τ as

$$\tau = 1 - \frac{1}{4n \log N},$$

which will be used to control the gap between s and s^τ in the analysis below. The graph construction process for the gap-preserving reduction is illustrated in Figure 1. For the constructed graph G' , we denote by $\text{OPT}(G')$ the maximum size of a **Flexi-clique**. **YES Case.** Assume that G contains a clique $Q \subseteq V$ with $|Q| = K$, and let $S = \bigcup_{v \in Q} C_v \subseteq V'$. Clearly, the set S has size KB , and the induced subgraph $G'[S]$ is connected since all cliques corresponding to nodes in Q are adjacent. Consider any node $x \in S$, since C_v is a clique, x has $B - 1$ neighbours inside $G'[C_v]$. Moreover, from the other $K - 1$ cliques in S , the node x has in total $(K - 1)B$ neighbours. Therefore,

$$\deg_{G'[S]}(x) \geq (B - 1) + (K - 1)B = KB - 1. \quad (1)$$

As $\tau \in (0, 1)$, $KB - 1 \geq \lfloor (KB)^\tau \rfloor$ for the chosen value of B . Hence $\deg_{G'[S]}(x) \geq \lfloor |S|^\tau \rfloor$ for all $x \in S$, and thus S is a valid **Flexi-clique** with $|S| = KB$, implying

$$\text{OPT}(G') \geq KB. \quad (2)$$

NO Case. Now assume that every clique in G has size at most K/c . Let $S \subseteq V'$ be an arbitrary **Flexi-clique** in G' . For each $v \in V$, define $a_v = |S \cap C_v|$, and let

$$U = \{v \in V : a_v > 0\}. \quad (3)$$

Denote

$$s = |S| = \sum_{v \in U} a_v. \quad (4)$$

We first bound the total number of non-neighbours that a node in a **Flexi-clique** can have. Recall that $\tau = 1 - \frac{1}{4n \log N}$ and that $s \leq nB = N$. Thus,

$$s - s^\tau \leq \frac{s}{4n} \leq \frac{B}{4}. \quad (5)$$

Fix any $u \in U$ and consider an arbitrary node $x \in S \cap C_u$. By construction of G' , the degree of x in the induced subgraph $G'[S]$ is

$$\deg_{G'[S]}(x) = (a_u - 1) + \sum_{\substack{v \in U: \\ (u,v) \in E}} a_v. \quad (6)$$

Therefore, the number of nodes in S that are non-adjacent to x is

$$(s - 1) - \deg_{G'[S]}(x) = \sum_{\substack{v \in U: \\ (u,v) \notin E}} a_v. \quad (7)$$

Since S is a **Flexi-clique**, we have $\deg_{G'[S]}(x) \geq \lfloor s^\tau \rfloor$, and hence

$$\sum_{\substack{v \in U: \\ (u,v) \notin E}} a_v \leq (s - 1) - \lfloor s^\tau \rfloor \leq s - s^\tau \leq \frac{B}{4}, \quad (8)$$

where the last inequality follows from (5). Now let C be a maximum clique in the induced graph $G[U]$. By the **NO**-case assumption, we have $|C| \leq K/c$. By maximality of C , for every $v \in U \setminus C$ there exists a node $u \in C$ such that $(u, v) \notin E$, and we assign each $v \in U \setminus C$ to one such u . For any fixed $u \in C$, it follows from (8) that

$$\sum_{v \in U \setminus C} a_v \leq \sum_{\substack{v \in U: \\ (u,v) \notin E}} a_v \leq \frac{B}{4}. \quad (9)$$

Summing over all $u \in C$, we obtain

$$\sum_{v \in U \setminus C} a_v \leq |C| \cdot \frac{B}{4}. \quad (10)$$

Since each block C_v has size exactly B , we also have

$$\sum_{u \in C} a_u \leq |C|B. \quad (11)$$

Combining this with (10), we conclude that

$$\begin{aligned} |S| &= \sum_{u \in C} a_u + \sum_{v \in U \setminus C} a_v \\ &\leq |C|B + |C| \cdot \frac{B}{4} = \frac{5}{4}|C|B \leq \frac{5}{4} \cdot \frac{K}{c}B. \end{aligned} \quad (12)$$

Therefore,

$$\text{OPT}(G') \leq \frac{5}{4} \cdot \frac{K}{c}B. \quad (13)$$

In conclusion, combining (2) and (13) yields a constant gap between **YES** and **NO** instances. Therefore, the **Maximum Flexi-clique** problem does not admit any constant-factor approximation unless $P = NP$. $\square$

B Complexity analysis

This section provides the degeneracy-based size bound and a conditional running time below 2^n for **EBA**. Throughout, $\tau \in (0, 1)$ is the parameter and $n = |V|$.

B.1 Size bound from degeneracy

LEMMA 3. *Let ω be the degeneracy of G , that is, the largest minimum degree over all subgraphs of G . Then any Flexi-clique $H \subseteq V$ satisfies $|H| \leq (\omega + 1)^{1/\tau}$.*

PROOF. By the definition of degeneracy, the induced subgraph $G[H]$ contains a node x with $d(x, G[H]) \leq \omega$. Since H is a Flexi-clique, $\lfloor |H|^\tau \rfloor \leq \delta(G[H]) \leq d(x, G[H]) \leq \omega$, hence $|H|^\tau < \omega + 1$, that is, $|H| < (\omega + 1)^{1/\tau}$. $\square$

B.2 A conditional bound below 2^n

Recall that EBA explores a branch-and-bound tree whose search nodes have the form $B = (S, C^r, C^{un}, D)$, and let $\mu(B) = |C^r| + |C^{un}|$ be the number of undecided candidates. In CP-Branching, the child B_i commits the candidates $v_1, \dots, v_i$ to S or D , so it reduces μ by at least i ; ordering the explored children by rank, the j -th of them reduces μ by at least j . While the worst-case running time is $O^*(2^n)$, the following shows that the width-based pivot drives the base strictly below 2 once the branching is kept narrow.

THEOREM 2. *Suppose the width-based pivot bounds the number of explored children at every search node to at most w . Then EBA runs in $O^*(\alpha_w^n)$, where α_w is the largest root of*

$$x^{w+1} - 2x^w + 1 = 0, \quad (14)$$

and $\alpha_w < 2$ for every $w \geq 2$.

PROOF. Let $T(\mu)$ bound the size of the subtree rooted at a search node of measure μ . Since at most w children are explored and the j -th reduces μ by at least j , we have $T(\mu) \leq \sum_{j=1}^w T(\mu - j)$. Substituting $T(\mu) = \alpha^\mu$ and dividing by $\alpha^{\mu-w}$ gives $\alpha^w = \sum_{l=0}^{w-1} \alpha^l = (\alpha^w - 1)/(\alpha - 1)$; multiplying by $(\alpha - 1)$ yields (14). Writing $f(x) = x^{w+1} - 2x^w + 1$, we have $f(1) = 0$ and $f(2) = 2^{w+1} - 2 \cdot 2^w + 1 = 1 > 0$, while f attains a negative local minimum at $x = 2w/(w+1) < 2$ for $w \geq 2$, so its largest root α_w lies in $(1, 2)$. As the per-node work is polynomial in the size of G , $T(n) = O^*(\alpha_w^n)$. $\square$

For instance, $w = 2, 3, 4$ give $\alpha_w \approx 1.618, 1.839, 1.928$, approaching 2 as w grows. This width bound is precisely what the width-based pivot targets. The pivot selects the node p of smallest margin $\bar{d}_{\text{margin}}(p)$, the number of further non-neighbours p can tolerate before violating its degree constraint, and orders the non-neighbours of p first over the whole reachable set. Each admitted non-neighbour raises the in- S non-degree of p by one, so after $\bar{d}_{\text{margin}}(p)$ of them p saturates and every later child, including the include-all child whose partial set then contains all non-neighbours of p , is pruned. The number of explored children is thereby held to about $\bar{d}_{\text{margin}}(p) + 1$, so w is on the order of the pivot's margin. On instances where this margin remains small, the base is well below 2, whereas the sublinear degree requirement of Flexi-clique prevents any constant w from being guaranteed in the worst case.

B.3 Absence of a size threshold for connectivity

The Flexi-clique model imposes only a size-dependent degree constraint and, unlike related dense-subgraph models, provides no size threshold beyond which connectivity is guaranteed. We make this precise below.

PROPERTY 1 (NO SIZE THRESHOLD FOR CONNECTIVITY). *For any $\tau \in (0, 1)$ and any integer h , there exists a node set $H \subseteq V$ with $|H| \geq h$ such that every node $v \in H$ satisfies $d(v, G[H]) \geq \lfloor |H|^\tau \rfloor$, yet $G[H]$ is disconnected.*

PROOF. Fix any $\tau \in (0, 1)$ and any integer h . Choose an integer s such that

$$2s \geq h \quad \text{and} \quad s - 1 \geq \lfloor (2s)^\tau \rfloor. \quad (15)$$

Table 1. Per-node branching width as graph size n grows (LFR, $\tau=0.7$), and on a real million-node graph.

Graph	n	W_{med}	W_{max}	αw_{med}
LFR	10^3	3	6	1.839
LFR	10^4	3	17	1.839
LFR	10^5	2	11	1.618
LFR	10^6	2	8	1.618
Florida	1.07×10^6	3	4	1.839

Such an s always exists for sufficiently large s , since $(2s)^\tau$ grows strictly sub-linearly in s for $\tau < 1$. Now construct a graph $G = (V, E)$ consisting of two disjoint cliques C_A and C_B , each of size s , with no edges between them, and let $H = C_A \cup C_B$, so that $|H| = 2s \geq h$. Each node $v \in H$ has degree $s - 1$ in $G[H]$, and by (15) we have $s - 1 \geq \lfloor (2s)^\tau \rfloor = \lfloor |H|^\tau \rfloor$; hence every node satisfies the Flexi-clique degree constraint. However, since there is no edge between C_A and C_B , the induced subgraph $G[H]$ is disconnected. Therefore, for any arbitrarily large size h , there exists a subgraph of size at least h that satisfies the Flexi-clique degree constraint but remains disconnected, so the Flexi-clique model admits no size-based lower bound that guarantees connectivity. $\square$

Consequently, connectivity cannot be inferred from the degree constraint and must be enforced explicitly during the search, which motivates the connectivity-aware branching of EBA.

C Deferred experiments

C.1 Empirical branching width

Theorem 2 shows that if the width-based pivot keeps the number of explored children at every search node bounded by a constant w , then EBA runs in $O^*(\alpha_w^n)$ with $\alpha_w < 2$. The bound is conditional: the sublinear degree requirement of Flexi-clique prevents any constant w from being guaranteed in the worst case. We therefore measure the per-node branching width w empirically by instrumenting EBA to record, at every CP-Branching search node, the number of children it actually explores.

The key question is whether w stays small relative to the graph size. Two findings show that it does. First, the width does not scale with n : Table 1 reports the width on a fixed LFR family as the number of nodes grows from 10^3 to 10^6 , and the median stays at 2–3 (effective base $\alpha_2=1.618$ to $\alpha_3=1.839$) across four orders of magnitude of graph size. Second, on real networks (Table 2) the width remains a tiny fraction of n —the maximum width is at most a few hundred even on graphs with hundreds of thousands to millions of nodes, i.e. $W_{\text{max}}/n \leq 0.8\%$ and as low as 4×10^{-6} on Florida. What governs the width is the degree structure, not the size: consistent with Lemma 3, w tracks the Flexi-clique (answer) size, which is bounded by the graph’s degeneracy rather than by n . Thus, while the worst-case constant cannot be guaranteed, in practice the branching width stays orders of magnitude below n and does not grow with it, so EBA effectively explores a search tree far narrower than the trivial 2^n bound.

Table 2. Branching width on real networks ($\tau \in \{0.5, 0.7\}$); $W_{\max}/n$ stays below 1%.

Dataset	n	ans.	W_{med}	W_{max}/n
Astrophysics	16,046	252	44	0.00798
Condensed	39,577	360	231	0.00644
Coauthorship	90,303	255	20	0.00194
Gowalla	196,591	1443	255	0.00130
DBLP	317,080	899	34	0.00080
Florida	1,070,376	7	3	3.7×10^{-6}

C.2 Contribution of connectivity-preserving branching

To isolate the value of CP-Branching, we compare it against two exact branching schemes that explore the same search space under identical pruning (the same degree and size rules and the same FPA lower bound) but without connectivity-by-construction: **SE**, a set-enumeration scheme in the style of quasi-clique enumerators that includes each candidate in turn and recurses on the strictly later ones; and **Naive**, binary include/exclude branching. Because both add candidates without regard to adjacency, the partial set may disconnect, so they must verify connectivity explicitly whenever a solution is taken, whereas CP-Branching never does. Table 3 reports the number of explored search-tree nodes; the comparison thus isolates the branching scheme itself.

CP-Branching explores dramatically fewer search nodes and is correspondingly faster—on Football at $\tau=0.7$ it finishes after 8198 nodes in 0.11 s, whereas set-enumeration explores 26.4M nodes and takes 30.1 s (280× slower), with the gap widening as graphs grow. **Naive** is competitive on the smallest instances but also blows up (e.g. 191,745 nodes on Football $\tau=0.7$). Across all instances CP-Branching performs zero explicit connectivity checks, since connectivity is preserved by construction. Both the search-node count and the running time confirm that the gain is attributable to the branching scheme rather than to pruning, which is held identical.

Table 3. Branching scheme comparison under identical pruning: explored nodes / time(s). CP-Branching = ours, SE = set-enumeration, Naive = binary include/exclude.

Graph	τ	CP-Branching	SE	Naive
Dolphin	0.7	276 / 0.005	2,555 / 0.012	447 / 0.004
Railway	0.7	212 / 0.011	48,342 / 0.10	6,087 / 0.039
Football	0.8	751 / 0.021	261,789 / 0.33	6,275 / 0.034
Football	0.7	8,198 / 0.11	26.4M / 30.1	191,745 / 0.60

C.3 Branching-order comparison

The connectivity-preserving scheme of Appendix C.2 fixes which candidates may be branched on; a further degree of freedom is the order in which they are taken. Our default is the margin pivot of Section ??, which branches on the least-slack vertex first. Table 4 compares it against a high-degree-first and a random ordering under identical pruning. The three orderings track each other closely on the easy instances, but the margin pivot is the most consistent: it is fastest at $\tau=0.3$ —the hardest setting, where the candidate set is largest—on almost every dataset, and remains the fastest, or within a few percent of the best, at $\tau=0.7$; the orderings are more mixed at $\tau=0.5$. Since the margin pivot never incurs a large penalty and most often yields the lowest running time, EBA adopts it as the default branching order.

Table 4. Branching-order running time (s); each cell is $\tau=0.3/0.5/0.7$, fastest in **bold**. Placeholder—pending root-fixed re-measurement.

Dataset	Margin	HighDeg	Random
Email	0.01/0.02/0.03	0.01/0.02/0.03	0.01/0.02/0.03
Erdős	0.04/0.03/0.03	0.04/0.03/0.03	0.04/0.03/ 0.02
PGP	0.23/0.04/0.03	0.28/0.06/0.03	0.27/0.06/0.03
Astrophysics	6.32/ 0.66/1.05	6.22/0.81/1.88	6.20/0.82/1.18
Dimacs	0.29/0.08/0.07	0.30/0.11/0.07	0.30/0.09/0.08
CAIDA	0.36/0.12/0.08	0.38/ 0.11/0.09	0.38/0.12/0.09
Condensed	8.84/4.19/0.34	8.97/ 2.12/0.42	9.02/2.73/0.41
Coauthorship	5.30/2.16/0.35	5.52/2.25/0.36	5.58/ 1.95/0.45
Brightkite	6.78/1.10/0.44	7.06/1.15/0.54	7.04/1.15/0.55
Caster	11.81/3.03/1.97	12.05/ 2.97/2.05	12.19/3.44/2.04
Flickr	92.01/51.20/28.94	93.13/52.34/ 28.54	93.14/ 50.21/28.57
Gowalla	90.86/45.61/3.25	92.88/ 23.00/3.09	98.56/27.78/ 3.08
DBLP	104.65/23.48/3.72	108.71/25.84/4.01	111.58/23.82/3.74
Florida	T/O/ 1.56/1.49	T/O/1.62/1.70	T/O/1.60/1.70
YouTube	429.60/58.09/T/O	442.17/63.13/T/O	444.91/64.22/T/O

C.4 Comparison with exact quasi-clique search (IterQC)

The closest exact baseline is IterQC, a state-of-the-art maximum γ -quasi-clique solver. Both target non-hereditary cohesive models, but the quasi-clique is fundamentally easier: its degree threshold grows linearly with the subgraph size, and for $\gamma \geq 0.5$ a quasi-clique is guaranteed to be connected with diameter at most two, so IterQC can restrict the search to local, bounded-diameter neighbourhoods and never needs an explicit connectivity mechanism. Flexi-clique, by contrast, has a sublinear threshold $\lfloor n^\tau \rfloor$ and requires connectivity explicitly, which is why EBA relies on connectivity-preserving branching. Moreover IterQC does not even apply for $\gamma < 0.5$ (it returns no result for $\gamma = 0.3$), whereas EBA solves the full range of τ .

We compare EBA (τ) against IterQC (γ) at the matched parameter values 0.5 and 0.7 under a 1-day budget (Tables 5 and 6). Two observations stand out. First, despite solving the harder problem, EBA is competitive and often far faster: e.g. at 0.5 it finishes Gowalla in 45 s versus IterQC’s 1007 s (22 $\times$) and YouTube in 56 s versus 7034 s (126 $\times$); on the instances where IterQC is faster the gap is small. Second, the maximum Flexi-clique is substantially larger than the maximum quasi-clique at the same parameter (e.g. Flickr at 0.5: 9024 vs. 1051; Astrophysics: 993 vs. 107), so EBA attains comparable running time while certifying optimality over a much larger and harder search space.

Table 5. Running time (s): EBA (τ) vs. IterQC (γ). 1-day timeout; IterQC has no 0.3 setting.

Dataset	param = 0.5		param = 0.7	
	EBA	IterQC	EBA	IterQC
Email	0.02	0.04	0.03	0.09
Erdős	0.03	0.03	0.03	0.05
PGP	0.05	2.10	0.03	0.18
Astrophysics	0.73	0.52	0.98	0.56
Dimacs	0.10	0.32	0.07	0.32
CAIDA	0.11	1.02	0.08	0.83
Condensed	4.19	0.55	0.29	0.55
Brightkite	1.26	0.65	0.51	0.65
Caster	3.23	T/O	1.94	3.08
Coauthorship	1.97	0.26	0.30	0.25
Flickr	45.92	4.03	26.65	32.69
Gowalla	45.08	1007.24	2.93	0.98
DBLP	22.93	1.15	3.15	0.60
Florida	1.47	1.18	1.58	1.32
YouTube	55.75	7034.40	T/O	4.38
Pokec	T/O	271.77	86.31	14.80
Wiki	T/O	T/O	245.62	T/O
LiveJournal	T/O	137.26	439.09	27.06

Table 6. Solution size: maximum Flexi-clique (EBA, τ) vs. maximum quasi-clique (IterQC, γ).

Dataset	param = 0.5		param = 0.7	
	EBA	IterQC	EBA	IterQC
Email	440	67	137	42
Erdős	80	21	30	12
PGP	194	49	116	42
Astrophysics	993	107	252	62
Dimacs	195	51	88	35
CAIDA	168	45	77	28
Condensed	360	49	93	31
Brightkite	561	101	208	72
Caster	1370	T/O	460	82
Coauthorship	255	45	87	43
Flickr	9024	1051	1846	576
Gowalla	1443	87	270	51
DBLP	899	115	259	114
Florida	15	7	7	5
YouTube	2208	75	T/O	39
Pokec	T/O	75	244	51
Wiki	T/O	T/O	638	T/O
LiveJournal	T/O	473	1456	390

References

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